

2018 Ph H2 Q13

Section: Our Dynamic Universe

Topic: Pressure in liquids, density from graph

(a) Calculate the pressure due to water of density $1.00 \times 10^3 \text{ kg m}^{-3}$ at a depth of 0.35 m.

(b) For another liquid, measurements of pressure at different depths are recorded.

(i) Plot a graph of p against h .

(ii) Calculate the gradient of the graph.

(iii) Determine the density of the liquid.

Worked solution

(a)

$$p = \rho gh.$$

$$= (1.00 \times 10^3)(9.8)(0.35) = 3430 \text{ Pa}.$$

Answer: $3.4 \times 10^3 \text{ Pa}$

(b)(i)

Plot of pressure (Pa) against depth (m) is a straight line through the origin, showing proportionality between p and h .

Answer: Straight line graph through origin

(b)(ii)

$$\text{Gradient} = \Delta p / \Delta h.$$

$$= (6200.0 - 1200.0) / (0.5 - 0.1).$$

$$= 12500.0 \text{ Pa m}^{-1}.$$

Answer: $1.25 \times 10^4 \text{ Pa m}^{-1}$

(b)(iii)

From formula $p = \rho gh$, slope = ρg .

$$\rho = \text{slope}/g = 12500.0/9.8 = 1275.5 \text{ kg m}^{-3}.$$

Answer: $1.28 \times 10^3 \text{ kg m}^{-3}$

Final answers

(a) $3.4 \times 10^3 \text{ Pa}$

(b)(i) Straight line graph through origin

(b)(ii) Gradient = $1.25 \times 10^4 \text{ Pa m}^{-1}$

(b)(iii) Density = $1.28 \times 10^3 \text{ kg m}^{-3}$

Revision tips

- Pressure in liquid: $p = \rho gh$.
- Graph of pressure vs depth gives straight line through origin.
- Gradient = ρg , so density = gradient/ g .
- Always convert kPa to Pa before calculations.
- Water has density $\approx 1000 \text{ kg/m}^3$; check if answer is realistic.